

Multisettings Training Lab

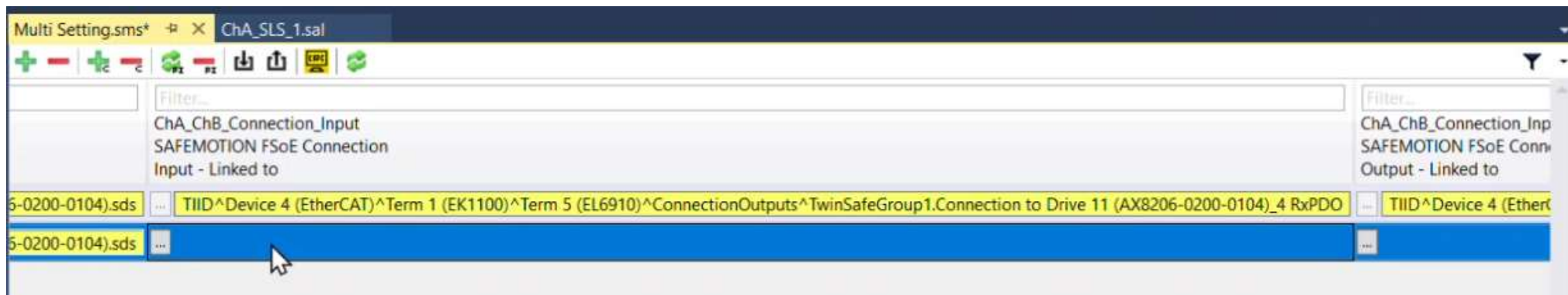
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- Introduction to Multi Settings.
- Navigate to the Multi Setting.sms file in the AX8000 safety project

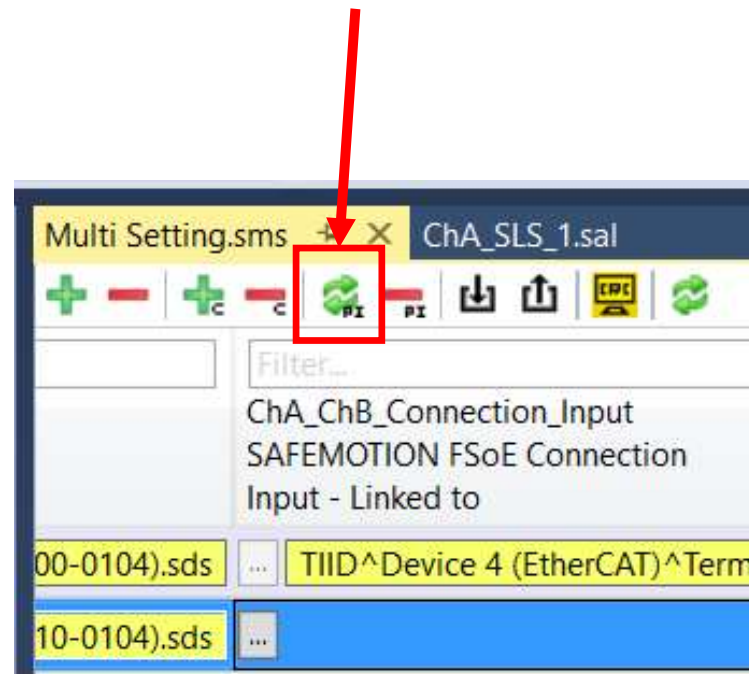
The screenshot displays the Beckhoff TwinCAT software interface. On the left, the 'Solution Explorer' pane shows a project hierarchy for 'Solution 'Multisettings' (2 projects)'. The hierarchy includes 'Multisettings', 'SYSTEM', 'MOTION', 'NC-Task 1 SAF', 'PLC', 'Untitled1', 'SAFETY', 'EL6910', 'AX8000', and 'Multi Setting.sms'. The 'Multi Setting.sms' file is selected and highlighted. On the right, the 'Multi Setting.sms' file is open, showing a table with columns for ID, Target System, Physical Device, Safe Address, and Alias Device. The table contains two rows: MS_1 and MS_2. MS_1 is associated with Drive 11 (AX8206-0200-0104) and has a safe address of 3. MS_2 is associated with Drive 10 (AX8206-0200-0104) and has a safe address of 513. The table also shows connections to TwinSafeGroup1 and Alias Devices.

ID	Target System	Physical Device	Safe Address	Alias Device
MS_1	AX891x	Drive 11 (AX8206-0200-0104)	3	EL6910^TwinSafeGroup1^Alias Devices^Connection to Drive 11 (AX8206-0200-0104).sds
MS_2	AX891x	Drive 10 (AX8206-0200-0104)	513	EL6910^TwinSafeGroup1^Alias Devices^Connection to Drive 10 (AX8206-0200-0104).sds

- Manually link FSOE data for our second drive.
 - See next slide

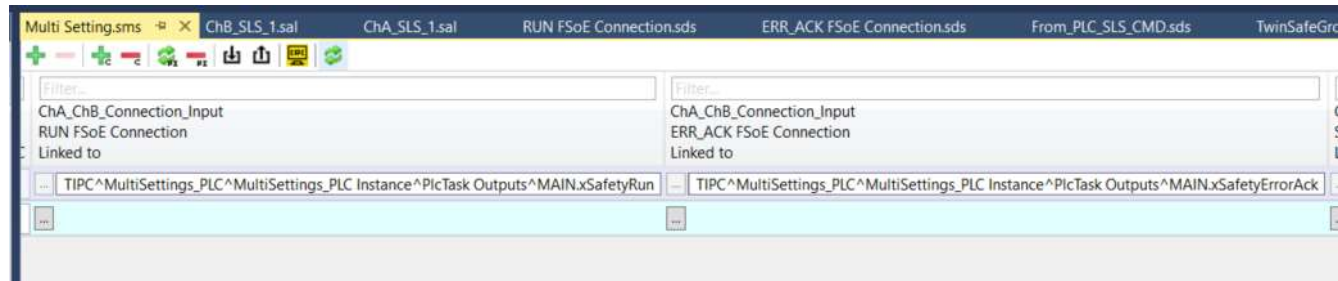
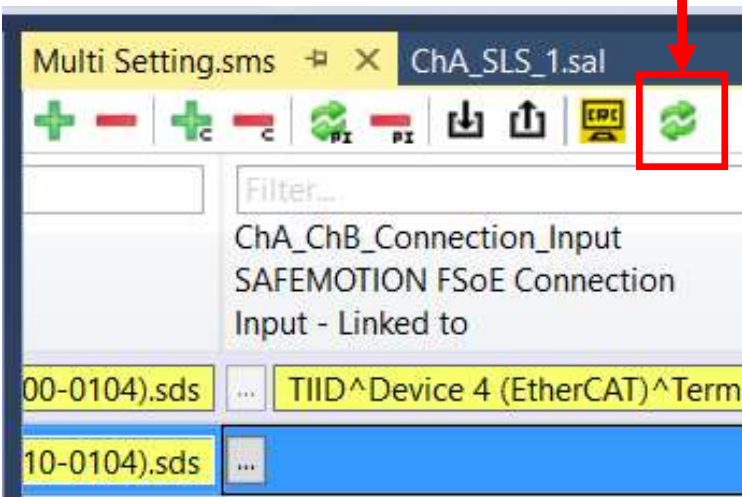


- Before we can link, we need to update the Process Image for the target system.
 - We do this with a simple button click below.

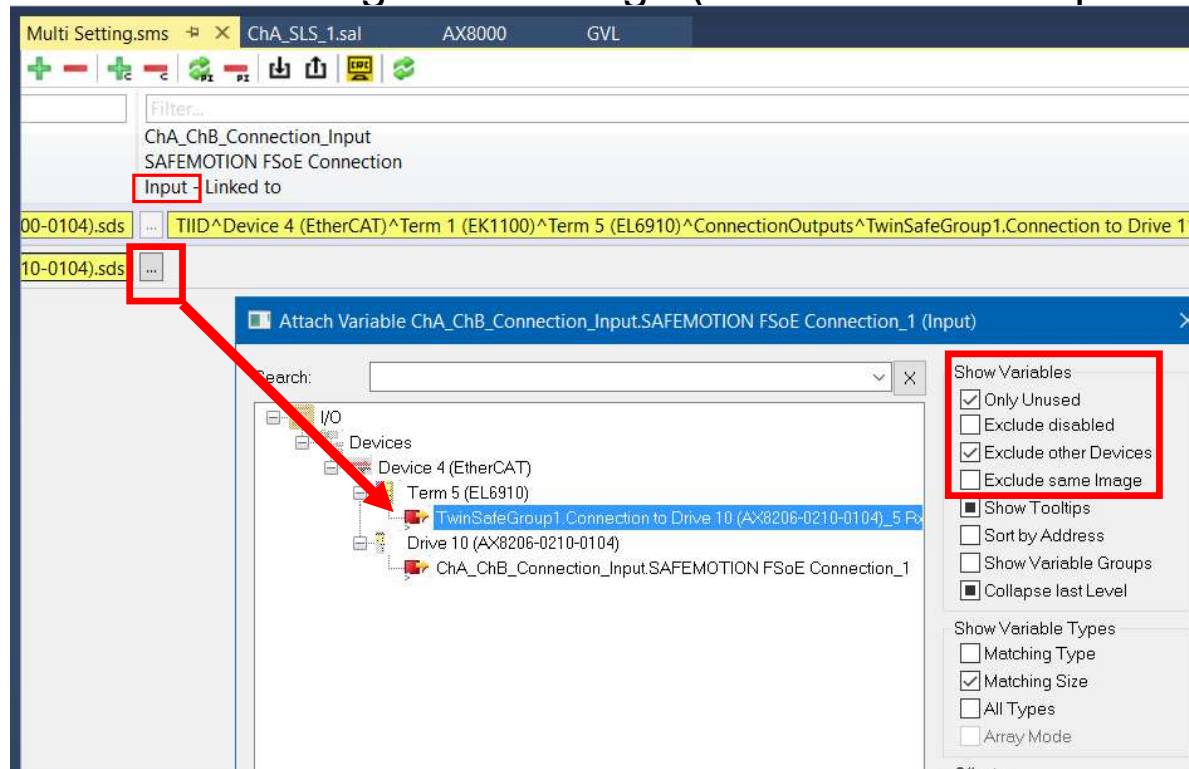


- Click the Refresh view button next

- This will update the Multi Setting.sms values for our SLS function blocks and RUN/ERR_ACK links



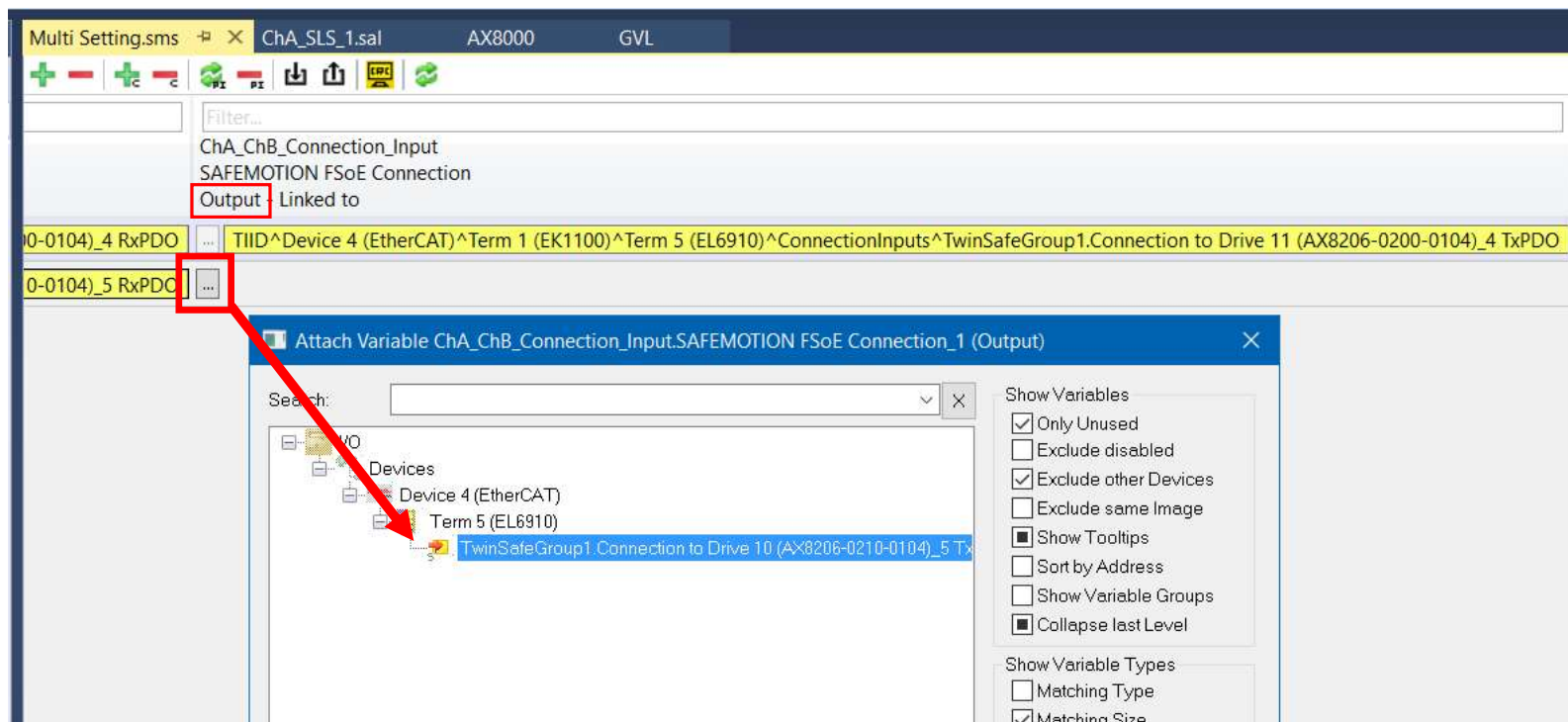
- Now, we can link the ChA_ChB_Connection_Input for the second drive.
 - SAFEMOTION FSoE Connection **Input**
 - Uncheck the box for Excluding Same Image (if no links show up in dialog).



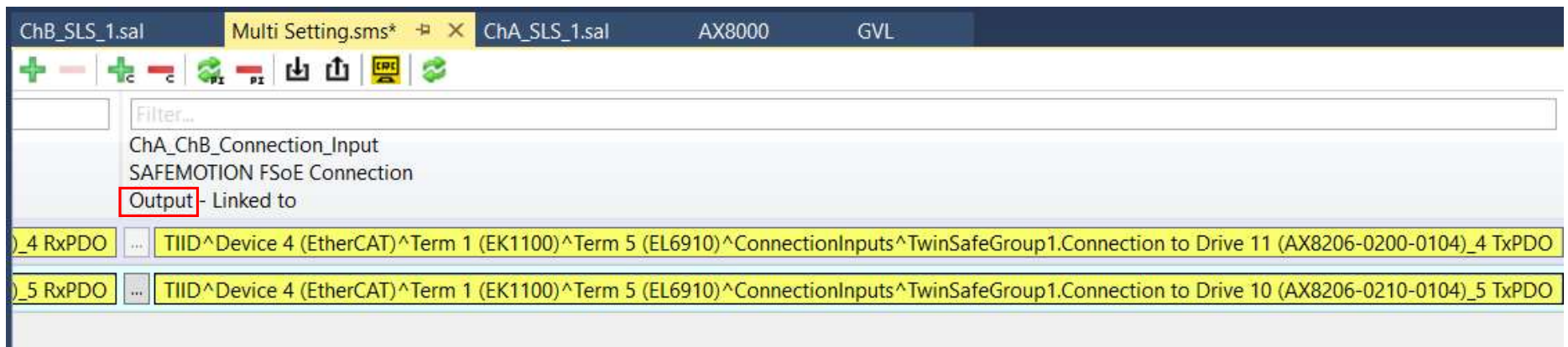
- Result should look like the image below.



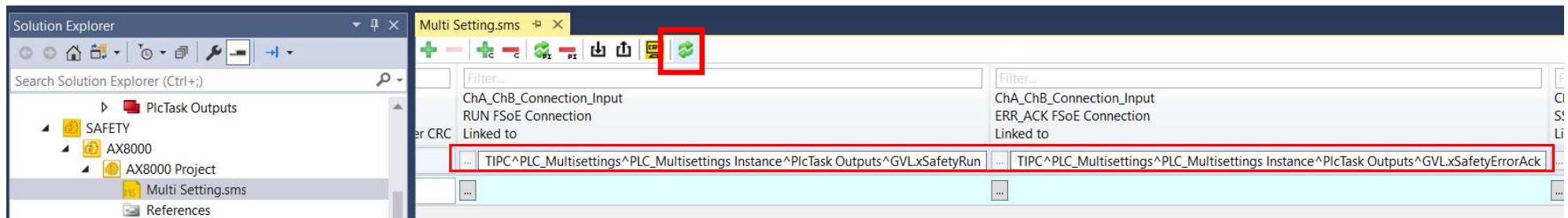
- Now, we can link the ChA_ChB_Connection_Input for the second drive.
 - SAFEMOTION FSoE Connection **Output**



- Result should look like the image below.



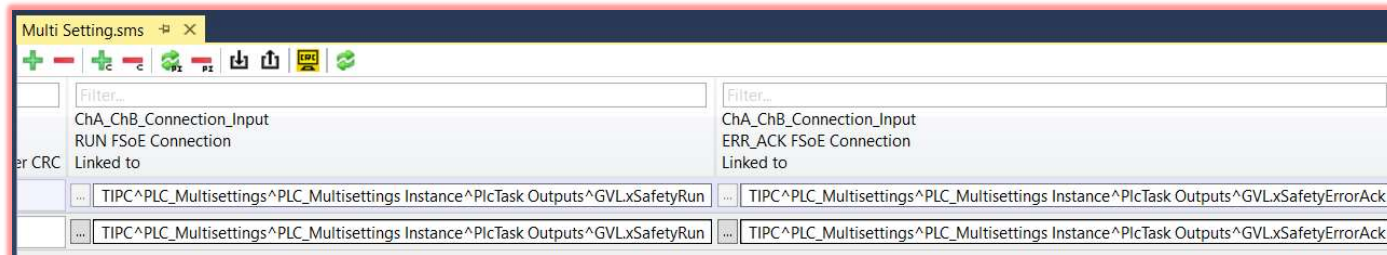
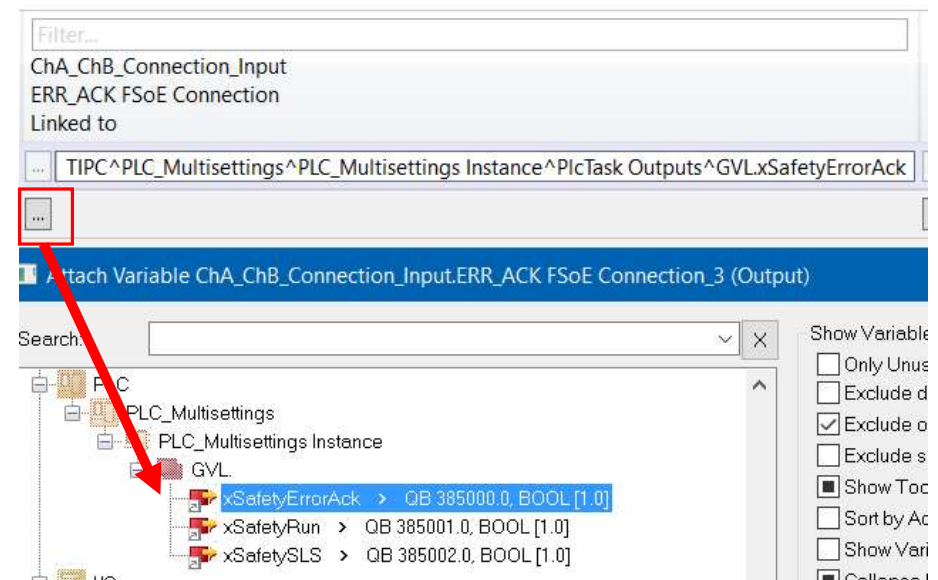
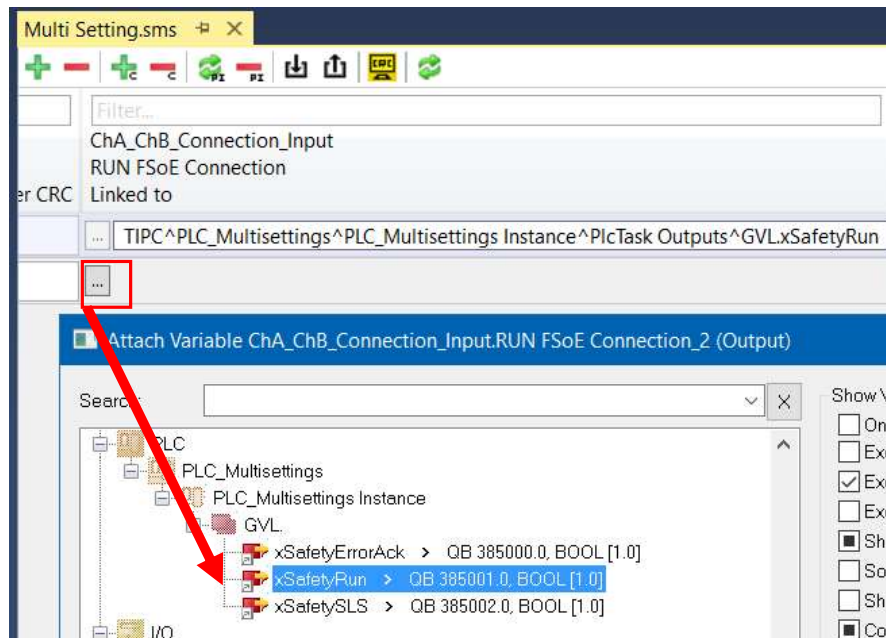
- Scroll to the right and you should see the RUN/ERR_ACK links.
 - Should resemble the image below. If you do not see a link already for the first drive, hit the Refresh view button.
 - If they still do not show up for the first drive, double check your Alias linking for the RUN/ERR_ACK in the AX8000 project and try hitting the Refresh button again.



Multisettings Training Lab

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- Link the RUN and ERR_ACK to our PLC variables.



- Scroll to the right.
- Input the SLS limit values for the second drive.

Multi Setting.sms								
	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
h_Output	ChA_ChB_Connection_Output	ChA_STO_SS1_ErrorHandling	ChA_SLS_1	ChA_SLS_1	ChB_STO_SS1_ErrorHandling	ChB_SLS_1	ChB_SLS_1	
	SS1_To_NC_ChB	SS1_ChA	SLS_1_Limit	SLS_1_Limit	SS1_ChB	SLS_1_Limit	SLS_1_Limit	
	Linked to	Delay Time (ms)	Minimum Value	Maximum Value	Delay Time (ms)	Minimum Value	Maximum Value	
	...	2000	-715828	715828	2000	-357914	357914	
	...	2000	0	0	2000	0	0	

Multi Setting.sms* ChA_SLS_1.sal MultiSettings_Demo MAIN [Online]								
	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
h_Output	ChA_ChB_Connection_Output	ChA_STO_SS1_ErrorHandling	ChA_SLS_1	ChA_SLS_1	ChB_STO_SS1_ErrorHandling	ChB_SLS_1	ChB_SLS_1	
	SS1_To_NC_ChB	SS1_ChA	SLS_1_Limit	SLS_1_Limit	SS1_ChB	SLS_1_Limit	SLS_1_Limit	
	Linked to	Delay Time (ms)	Minimum Value	Maximum Value	Delay Time (ms)	Minimum Value	Maximum Value	
	...	2000	-715828	715828	2000	-357914	357914	
	...	2000	-3006478	3006478	2000	-357914	357914	

- Placing a different value for Drive 10's ChA_SLS_1 Min/Max.

put	ChA_STO_SS1_ErrorHandling	ChA_SLS_1	SLS_1_Limit	SLS_1_Limit
	SS1_ChA		Minimum Value	Maximum Value
2000			-715828	715828
2000			-3006478	3006478

FeedConstant	60	mm/rev	
Velocity Resolution	4294967296	= 2^32	1 rev/ms
Safe Speed*	42	mm/sec	
SafeLimit Max**	3006478	INC/ms	
SafeLimit Min**	-3006478	INC/ms	
1000 * FeedConstant	60000	Calculation from Feedconstant	
Increments per Rev/ms	71582.7883	Calculation from Feedconstant	

- Save-All
- Multi-Download safety projects
- Activate Configuration.

- We are ready to test.